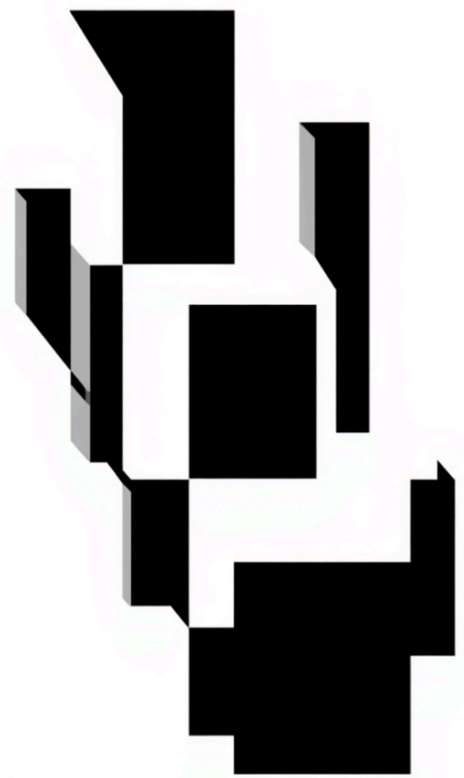
REVIEW -1

Project 34: Online Course Database Management System



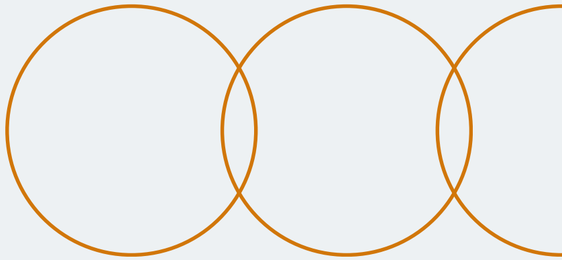
By Joy Peswani
25WU0102110

The Problem

Unstructured data significantly **impedes learning progress**:

- Students struggle to track their performance.
- Quiz attempts are often inaccurate or lost.
- Learning pathways lack organization and clarity.

Without a centralized database, instructors face difficulties in assessing student progress effectively. This disorganization can lead to frustration and disengagement among learners, as they cannot see their achievements or areas needing improvement. Implementing a structured database will enhance tracking, facilitate better feedback, and ensure that each learner's educational journey is both clear and measurable.

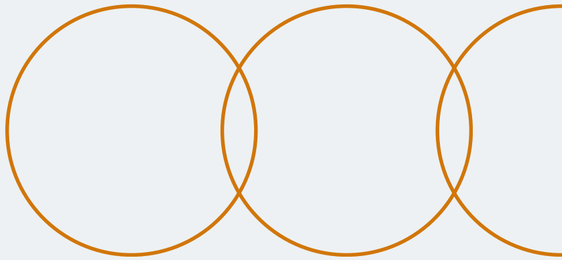


Scope

The objective of the project is to design a **normalized 3NF relational database** that addresses the needs of the online course management system.
The key goals include:

- Track the full **educational lifecycle** of users.
- Automate attempt tracking for quizzes and assignments.
- Streamline the issuance of certificates upon course completion.

By implementing these functionalities, we aim to enhance user experience and ensure accurate data management throughout the learning process, ultimately improving educational outcomes for both learners and instructors.

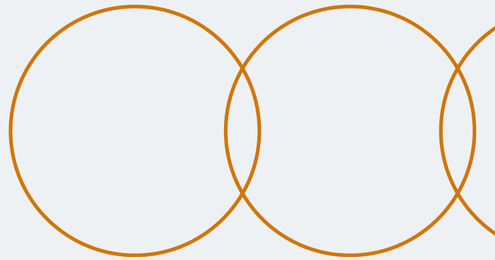


Users & Roles

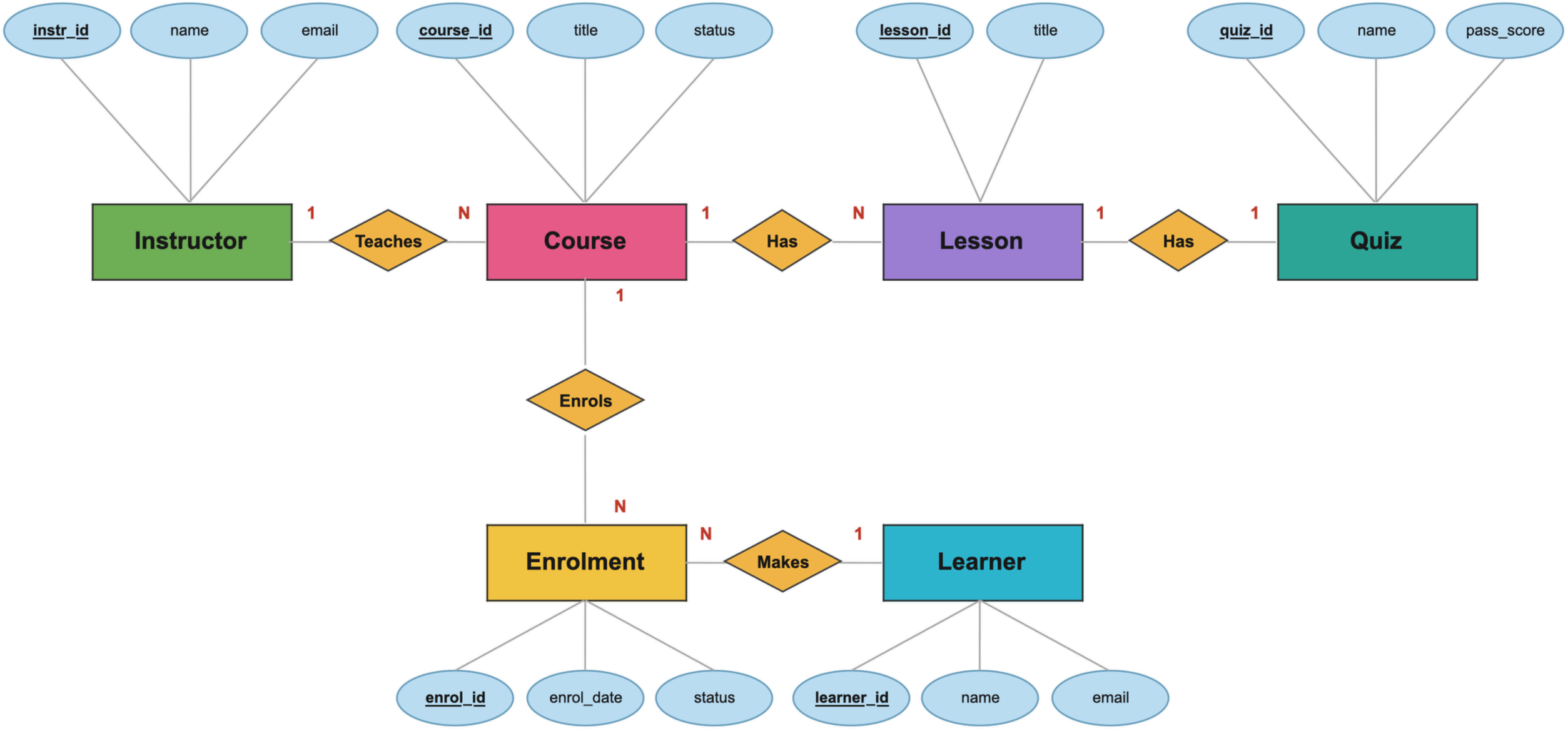
The system incorporates three key user roles:

- **Instructors**: Responsible for publishing courses and providing educational content.
- **Learners**: Users who enroll in courses, take quizzes, and track their learning progress.
- **Administrators**: Monitor metrics and oversee the overall functionality of the system to ensure a smooth learning experience.

Each role plays a critical part in the educational ecosystem, enhancing user engagement and maintaining the integrity of the online course management system. This structure enables effective tracking and management of learning progress.



ER Diagram


instr_id
name
email
course_id
title
status
lesson_id
title
quiz_id
name
pass_score
Instructor
1
Teaches
N
Course
1
Has
N
Lesson
1
Has
1
Quiz
1
Enrols
N
Enrolment
N
Makes
1
Learner
enrol_id
enrol_date
status
learner_id
name
email

Database Schema

Relational Schema (3NF)

Instructor

instr_id (PK)
name
email

Course

course_id (PK)
title
status
instr_id (FK)

Lesson

lesson_id (PK)
title
course_id (FK)

Quiz

quiz_id (PK)
name
pass_score
lesson_id (FK)

Learner

learner_id (PK)
name
email

Enrolment

enrol_id (PK)
learner_id (FK)
course_id (FK)
enrol_date
status

Lesson_Progress

progress_id (PK)
enrol_id (FK)
lesson_id (FK)
status

Attempt

attempt_id (PK)
learner_id (FK)
quiz_id (FK)
score
attempt_no



Tracking Tables

User Progress and Actions Overview

Enrolment

The **Enrolment table** records each instance of a learner joining a course, with a unique identifier (PK: EnrolmentID) and references to both the learner and course (FKs).

Lesson Progress

The **Lesson Progress table** tracks individual learner advancements through course material, using a unique identifier (PK: ProgressID) and linking back to the enrolment data (FK: EnrolmentID).

Attempts

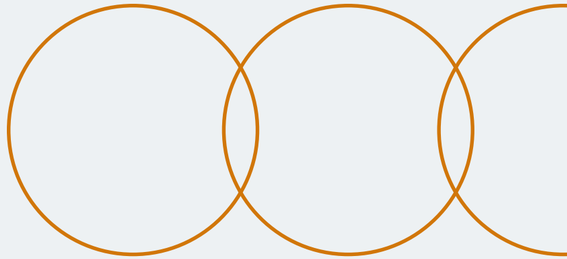
The **Attempt table** captures each learner's quiz attempts, uniquely identified (PK: AttemptID) and linked to the learner and specific quiz (FKs), tracking performance and engagement.

Functional Requirements

The system must be capable of:

- Supporting course publishing by instructors
- Tracking live lesson completion for learners
- Accurately calculating final scores based on quiz results
- Generating certificates upon course completion

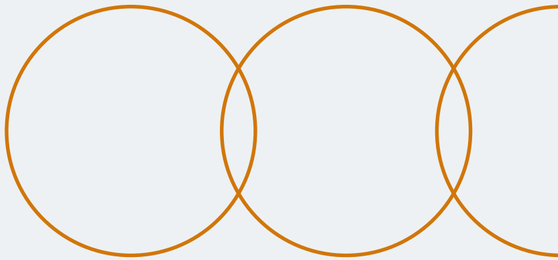
These capabilities are crucial for ensuring an effective learning experience. By automating these processes, the system enhances operational efficiency and provides users with timely insights into their progress. This integrated approach not only supports instructors and learners but also helps administrators monitor metrics effectively, fostering a streamlined educational environment.



Business Rules

- Limit users to **one enrolment** per course.
- Enforce progress should remain between **0-100**.
- Implement hard caps on **quiz attempts**.
- Require valid scores for all assessments.

These rules ensure a structured and fair learning environment, maintaining the integrity of the educational process. Each learner has a unique path, minimizing confusion and maximizing engagement. By enforcing these standards, the system promotes a clear framework for tracking progress and performance, aligning with the objectives of the Online Course DB initiative.



Conclusion

This project focuses on the design and implementation of a robust relational database management system for an Online Course Learning Progress Management System. Because unstructured data makes progress tracking unreliable, this system provides a centralized way to accurately manage content hierarchy, student enrolments, lesson progress, quiz attempts, final scores, and certificates. By translating a comprehensive Entity-Relationship model into a relational schema normalized up to Third Normal Form (3NF), the database guarantees data integrity and eliminates redundancy. Ultimately, this foundational structure will be implemented using MySQL, PostgreSQL, or Oracle and connected to a custom Java or Python front-end application to seamlessly support the entire e-learning lifecycle, from initial course publishing to automated certificate issuance